

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (CE)

January 2014

CE702 ADVANCED COMPUTER NETWORK

Date: 01.01.2014, Wednesday

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

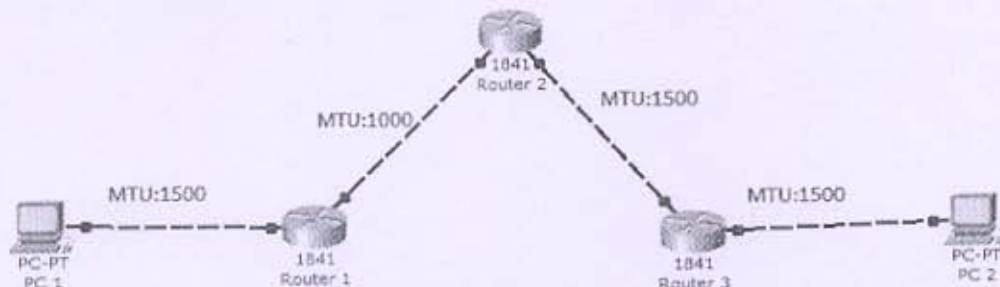
SECTION-I

Q-1 Answer the following questions.

- [A] With respect to TCP/IP model, _____ Layer is responsible for Flow Control. [1]
- [B] Which one of the following is correct statement [1]
- (i) Only data in datagram is fragmented at network layer.
- (ii) Packet including header is fragmented at network layer.
- [C] List error-reporting messages generated by ICMP. [1]
- [D] A host with IP address 137.23.56.23/16 sends packet to host with 137.23.67.9/16. Is the delivery is direct or Indirect? [1]
- [E] A 100-byte message is sent through a private internet using the TCP/IP protocol suite. If the protocol adds a 10-byte header at each layer, what is the efficiency of the system? (Efficiency: the ratio of the number of useful bytes to the number of total bytes) [1]
- [F] The Length of Physical address _____ bits, length of Logical Address is _____ bits and the length of Port Address is _____ bits. [1]
- [G] Which one of the following device forwards the error which is present in signal? [1]
- (1) Bridge (3) Repeater
- (2) Switch (4) Router
- [H] What do you mean by Service Point Addressing with respect to Transport Layer? [1]
- [I] What is limitation of Flooding routing protocol in multicasting? [1]
- [J] State True/False & Justify: ICMP sends error message to the previous Router. [1]
- [K] What is importance of Source Quench message? [1]

Q-2 Answer the following questions.

- [A] Compare and Contrast: CSMA/CD and CSMA/CA. [3]
- [B] What is method to recognize a class in IPv4 address? Also write down total number of host and network available in each class. [4]
- [C] By considering following network topology, explain fragmentation and reassembly procedure for sending of 7 MB of data from PC1 to PC2. [5]



OR

- [A] An address in a block is given as 180.8.17.9/20. Find the number of addresses in the [3]

block, the first address and last address.

- [B] By taking suitable block of IPv4 addresses, explain what is classfull and classless addressing. [4]
- [C] Explain the working of RIP protocol. [5]

Q-3 Answer the following questions.(Attempt any Two) [12]

- [A] How did private addresses solve problem of deficiency of IPv4 addresses when everyone wanted to access internet? Write down mechanism of solution.
- [B] Write and explain fields of IPv4 Header format.
- [C] Why do we require Multicast Routing Protocol, although we do have Routing Protocols (Like RIP, OSPF)?

SECTION-II

Q-4 Answer the following questions. [11]

- [A] "Developer cannot use bind system call into client program" Write your comment. [1]
- [B] What is unicasted multicasting? [1]
- [C] List the disadvantages of UDP protocol. [1]
- [D] Write down usage of exponential backoff algorithm in TCP. [2]
- [E] Give any scenario in UDP where ICMP protocol sends port unreachable message to requestor. [2]
- [F] Write down the procedure of encapsulating in TCP. [2]
- [G] List the applications in which UDP can be used? Why? [2]

Q-5 Answer the following questions. [12]

- [A] Write down meaning of various fields of composition of IPv6 address. [3]
- [B] By drawing appropriate state transition diagram, explain simultaneous close and denying a connection. [4]
- [C] How does UDP handle receiving and sending of packets? Which are the internal components of UDP which takes care of that activity? [5]

OR

- [A] Justify: TCP does not consider RTT of a retransmitted segment in its calculation of a new RTO. [3]
- [B] Which layer of TCP/IP model makes use of Maximum Transmission Unit? Why? [4]
- [C] What is silly windows syndrome problem? Write down the solutions of it. [5]

Q-6 Answer the following questions.(Attempt any Two) [12]

- [A] Draw and Explain TCP State Transition Diagram.
- [B] What are the Timer used in TCP?
- [C] Write and explain: Byte ordering functions in Socket Programming.